

Cover letter — Social Indicators Research submission

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The Editor-in-Chief
Social Indicators Research

Dear Editor,

We are pleased to submit our manuscript, **“Quantifying Hidden Exploitation: Dual-Method Prevalence Estimates of Labour Exploitation among Domestic Workers in the United Kingdom”**, for consideration in *Social Indicators Research*.

The paper responds to a persistent gap in the SDG 8.7 monitoring architecture: the populations most affected by labour exploitation — undocumented migrants, live-in domestic workers, tied-visa workers, agricultural day labourers — are systematically absent from the household and labour-force surveys that produce official social indicators. Existing prevalence estimates for modern slavery therefore rely on multiple-systems estimation from administrative case data, or on qualitative and civil-society case documentation. Neither approach yields the population-representative estimates that the SDG 8.7 target requires. The paper develops and demonstrates a replicable measurement design that addresses this gap through matched ego/alter survey instrumentation for hidden populations reachable through civil-society networks.

Two contributions justify submission to *Social Indicators Research* specifically. First, we apply respondent-driven sampling (RDS) and the network scale-up method (NSUM) to the same survey instrument administered to the same 85 respondents, eliciting ego-based and alter-based estimates of an identical set of five ILO-aligned indicators. To our knowledge this is among the first applications of a dual RDS + NSUM design to modern-slavery prevalence in any developed-economy setting. The design produces two complementary estimands — own-experience prevalence from RDS, network-visible prevalence from NSUM — and where they diverge, the gap is itself diagnostic evidence about which exploitation behaviours are observable across workers’ personal networks and which remain concealed. This “visibility

gradient” is a substantive social-indicator finding in its own right and has direct policy implications for network-mediated outreach programmes.

Second, we operationalise modern slavery as a graded formative indicator framework rather than a binary condition, drawing on the ILO’s own eleven-indicator framework and its associated survey-guidelines handbook. The design supports both indicator-level prevalence estimation for the five binary measures and a continuous formative-additive composite risk index across all eleven ILO indicators. The composite is anchored on the ILO severity ordering and is shown to be essentially weight-invariant under $\pm 50\%$ perturbation of the sub-indicator weights (Spearman $\rho = 0.992$). This gives *SIR* readers a measurement design that generalises to other hidden populations relevant to SDG 8.7 monitoring, with the substantive UK migrant-domestic-worker case as a hard, realistic testbed rather than as the endpoint of the paper.

We are careful throughout to characterise what the design does and does not license. The order-of-magnitude divergence between the RDS (35–84%) and MBSU NSUM Moderate (1–3%) estimates is interpreted as diagnostic rather than as competing estimates of a single quantity, and the paper explicitly does *not* claim a definitive national prevalence figure. Robustness of the substantive ordering of indicators is documented across four methodologically unrelated estimators (RDS-I, RDS-SS, MA-RDS, MBSU NSUM), across four assumed population sizes (50K, 100K, 980K, 1.74M), and against a leave-one-out exclusion of the single longest recruitment chain (seed #55, 12 respondents). All sensitivity analyses are reported in the Online Appendix.

Declarations.

- The manuscript is our original work and has not been published, is not under review, and is not being considered for publication elsewhere.
- All authors have read and approved the manuscript and agree to its submission to *Social Indicators Research*.
- Both authors declare no competing financial or non-financial interests.
- Ethics approval was obtained from the University of Nottingham Business School Research Ethics Committee (Ethics Review Application 202223021, granted January 2023) prior to fieldwork.
- Anonymised respondent-level data and all analysis code are archived on the Open Science Framework (view-only URL provided on the separate title page); the underlying repository will be made public on acceptance.
- We have followed the journal’s double-anonymous peer-review requirements: author names, affiliations, funding sources, and other potentially identifying material have been redacted from the main manuscript, the online appendix, and their figures. A separate anonymised title page, cover letter, and set of files have been prepared according to the *SIR* portal’s specifications.

Suggested reviewers. In the submission portal we suggest three reviewers whose expertise spans the paper’s methodological and substantive contributions: Dennis M. Feehan (UC Berkeley, network scale-up methodology); Sheldon X. Zhang (UMass Lowell, human-trafficking prevalence estimation); and Bridget Anderson (University of Bristol, migrant

domestic work and low-waged labour). Full rationales are given in the reviewer-suggestions section of the portal. We ask that colleagues at the University of Nottingham and members of our three participating civil-society organisations (Kalayaan, Voice of Domestic Workers, and the Latin American Women’s Rights Service) not be invited to review, on institutional-conflict and partnership-conflict grounds.

Length. The main manuscript is within the *SIR* 5,000–10,000 word limit including references. The Online Appendix (~ 20,000 words) documents the codebook, survey instrument, three-step bootstrap procedure, and eight sensitivity analyses. It is submitted as a single supplementary PDF and is not counted toward the word limit.

We thank you for considering the manuscript and look forward to the reviewers’ feedback.

Yours sincerely,

Scott Moser

On behalf of the authors